Guiding Diffusion Models to Reconstruct Flow Fields from Sparsa Data

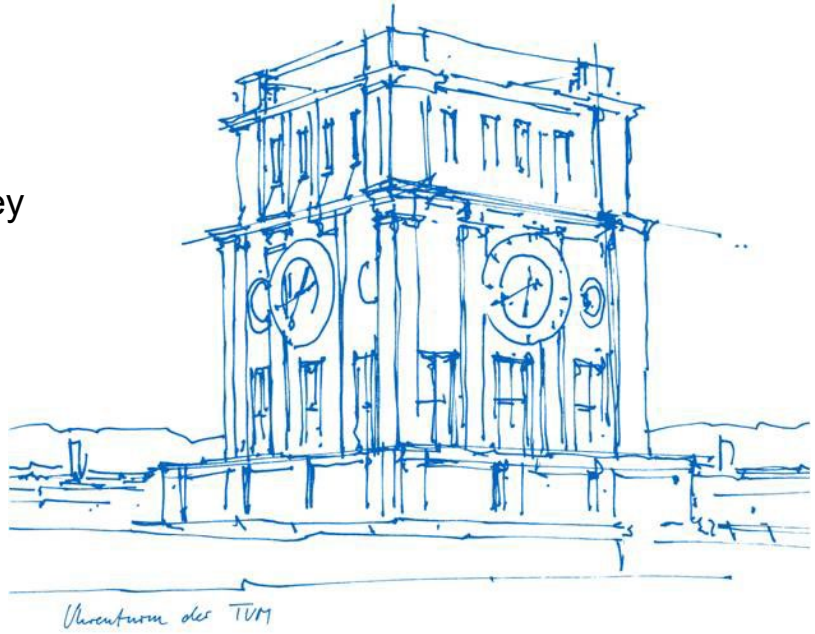
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Technische Universität München

17th WCCM & 10th ECCOMAS

Munich, 20.07.2026

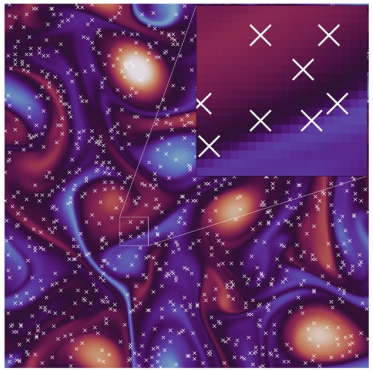


Motivation and problem formulation

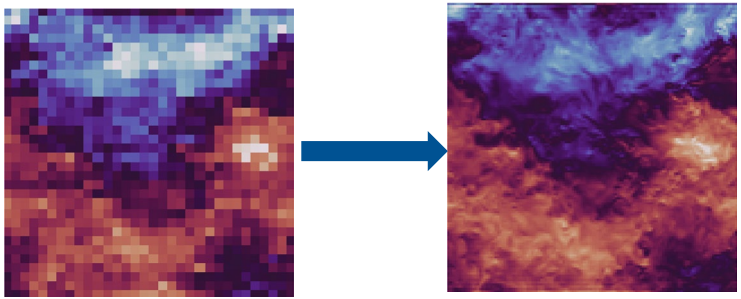
- Sparse data reconstruction
- Experimental fluid dynamics

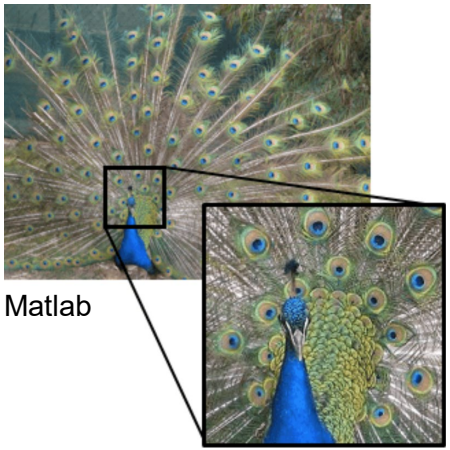


NASA



- Super-Resolution
- Computational fluid dynamics





Matlab

Motivation and problem formulation

Our goal

$$v_{LR} \xrightarrow{model} v_{HR}$$

Model?

- Classical interpolation: nearest, linear, cubic…
- Deep Learning:
  - Single-step NN supervised regression $\hat{x}_{HR}^{\theta} = NN_{\theta}(x_{LR})$ $\min_{\theta} ||x_{HR}^{\theta} - \hat{x}_{HR}||$
  - **Generative diffusion models → Masked Diffusion**

Motivation and problem formulation

Goals and Contributions

- Extensive benchmarks for **masked diffusion**
  - 2D Kolmogorov flow: 256^2
  - 2D weather data (ICON): 320^2 → 6.3 Km
  - 3D isotropic turbulence (JHTDB): 128^3
  - 3D channel flow: 192x64x192

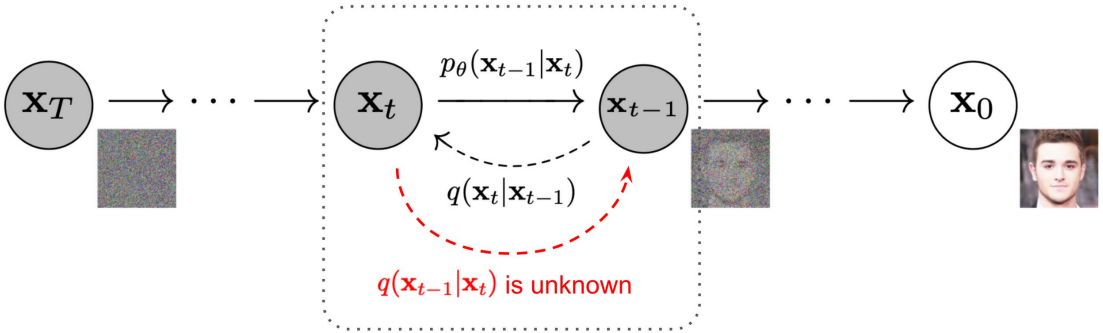
- Physical constraints
  - Weak → ConFIG

Methodology

Generative modeling

J. Ho, A. Jain, and P. Abbeel. Denoising Diffusion Probabilistic Models. 2020

Denoising Diffusion Probabilistic Models (DDPM)



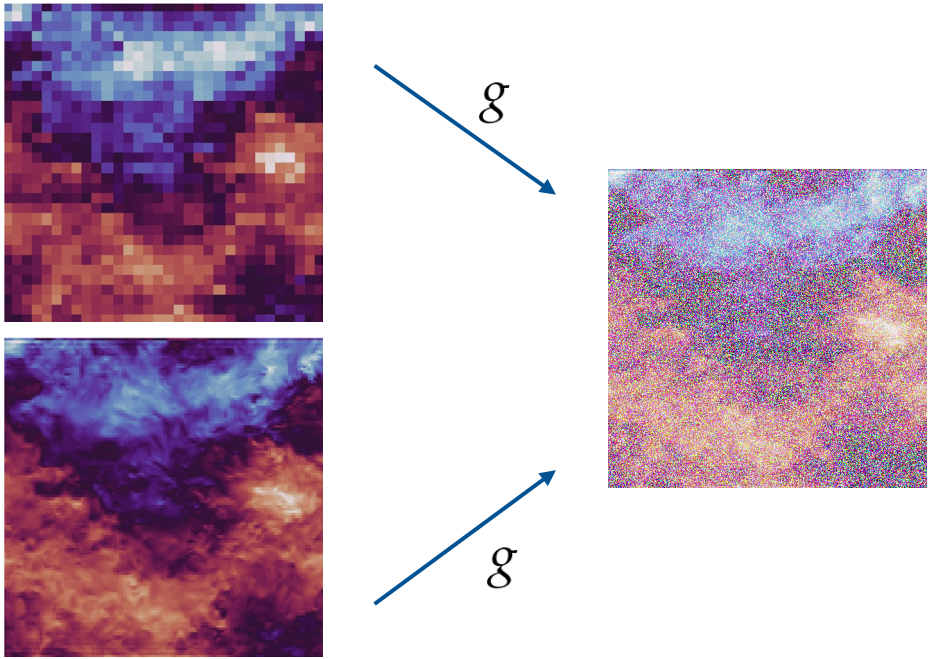
$$x_0^{(t)} = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(t, x_t)}{\sqrt{\bar{\alpha}_t}}$$

Prediction of the final sample

Methodology

Masked diffusion

If we degrade both HR and LR with gaussian noise…



$$g(x_{LR}) \approx g(x_{HR})$$

General training

- Train unconditional HR generative model
- Use LR to guide sampling

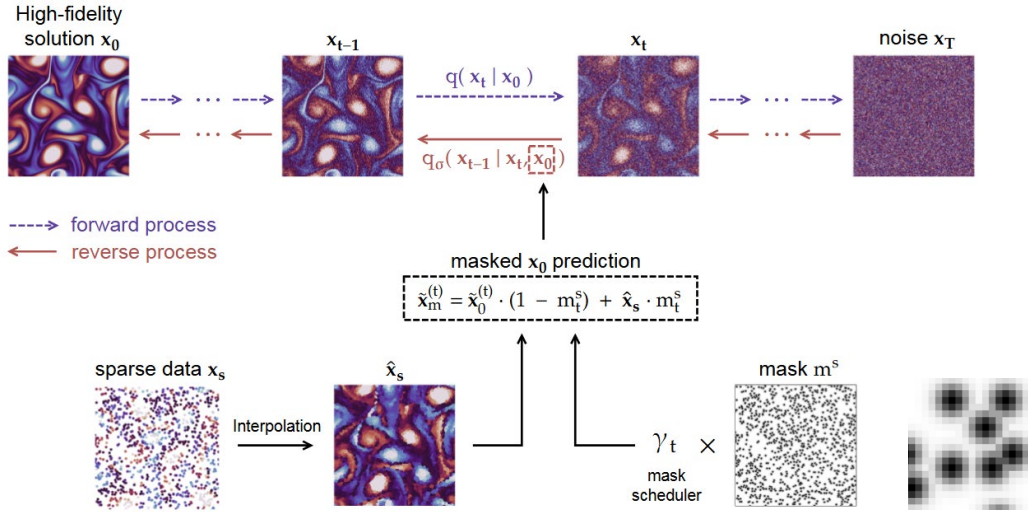
Methodology

Masked diffusion

Guiding the diffusion sampling

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_0^{(t)} + \sqrt{1-\bar{\alpha}_{t-1}} \epsilon_\theta(t, x_t)$$

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_g^{(t)} + \sqrt{1-\bar{\alpha}_{t-1}} \epsilon_\theta(t, x_t)$$



Combination of $x_0^{(t)}$ and x_{LR}

- Linear combination: $x_g^{(t)} = x_0^{(t)}(1-t) + \hat{x}_{LR} t$
- Binary mask: $x_g^{(t)} = x_0^{(t)} \cdot (1-m_t) + \hat{x}_{LR} \cdot m_t$
- Diffuse/Gaussian mask: $x_g^{(t)} = x_0^{(t)} \cdot (1-m_t) + \hat{x}_{LR} \cdot m_t$

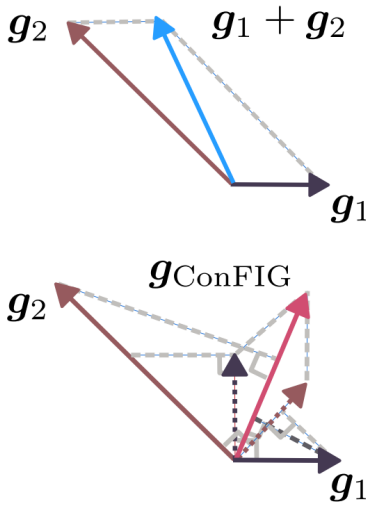
Methodology

Physics-Informed methods

Enforce $\nabla \cdot \vec{u} = 0$ or Navier-Stokes equations

- Weak constraint
  - Joint loss terms $L_{PIFM} = \mathbb{E}_{t,q(x_1),p_0(x_0)} \left(||u^{\theta}(x_t, t) - (x_1 - x_0)||^2 + \lambda ||\mathcal{R}(x_1^{(t)})||_2 \right)$
  - Non-conflicting gradients training (ConFIG)

Q. Liu, M. Chu, and N. Thuerey. ConFIG: Towards Conflict-free Training of Physics Informed Neural Networks. 2025



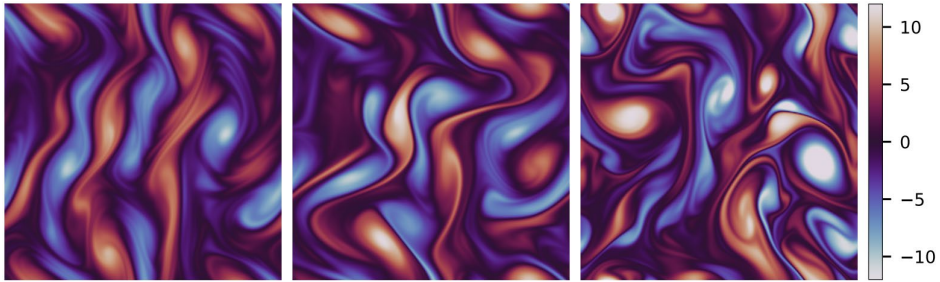
Results

Test cases

Kevin Debeire, Aytac Pacal, Pierre Gentine, Luis Medrano-Navarro, Nils Thuerey, and Veronika Eyring. “Physics-Constrained Adaptive Flow Matching for Climate Downscaling”. 2026

2D Kolmogorov flow

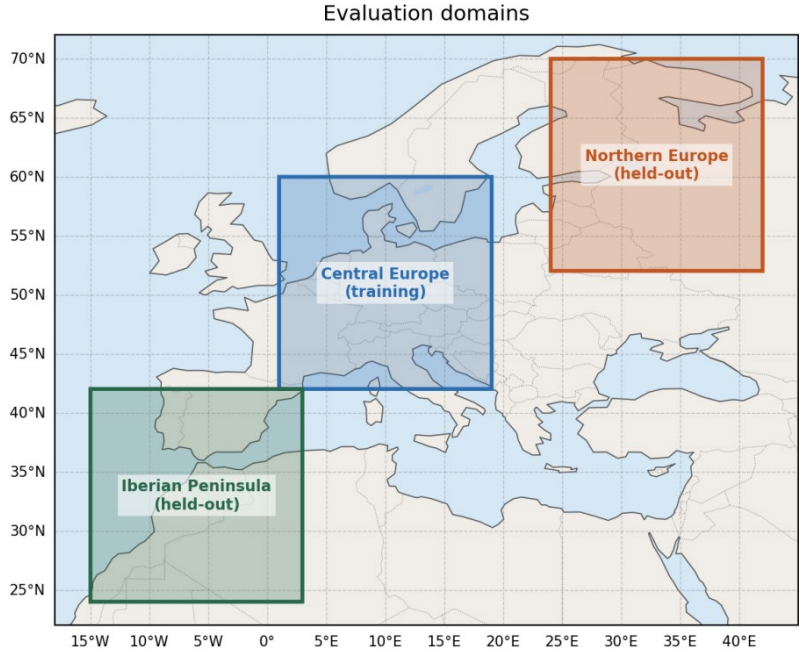
- $(0, 2\pi)^2$ with 256^2



D. Shu, Z. Li, and A. Barati Farimani. “A physics-informed diffusion model for high-fidelity flow field reconstruction”

2D weather data (ICON)

- 32x32 → 320x320



Results

Test cases

3D isotropic turbulence (JHTDB)

- $(0, 2\pi)^3$ with 128^3 grid points

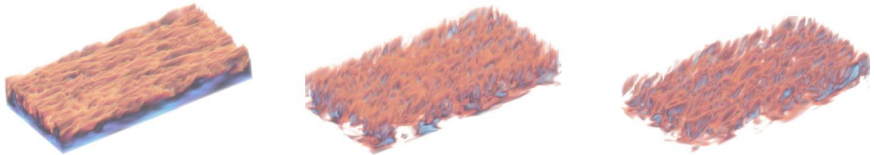


(a) x-velocity (b) y-velocity (c) z-velocity

J. H. T. Databases. Forced isotropic turbulence. https://turbulence.pha.jhu.edu/Forced_isotropic_turbulence.aspx

3D (half) **channel flow**

- $(0, 2\pi) \times (0,1) \times (0, \pi)$
- 192 x 64 x 192 stretched grid



(a) x-velocity (b) y-velocity (c) z-velocity

M. Chevalier, P. Schlatter, A. Lundbladh, and D. S. Henningson. “SIMSON : A PseudoSpectral Solver for Incompressible Boundary Layer Flows”

Results

Physics awareness

Test case	Training method	L_{res}
2D Kolmogorov flow	Std. Diffusion	1.27 ± 0.12
	PIDM_{dyn} [69]	$\mathbf{0.99 \pm 0.21}$
	ConFIG [71]	1.24 ± 0.32
	ConFIG_{u}	$\mathbf{0.68 \pm 0.15}$
3D isotropic turbulence	Std. Diffusion	1.03 ± 0.07
	PIDM_{dyn} [69]	1.08 ± 0.07
	ConFIG [71]	$\mathbf{1.01 \pm 0.05}$
	ConFIG_{u}	$\mathbf{0.80 \pm 0.05}$

Results

Numerical results

Evaluated on the known points

Physical residual

Learned similarity

Models	RMSE	P-RMSE	L_{res}	LSiM	E.spectrum
Supervised baseline	1.06 ± 0.16	0.81 ± 0.02	**0.21 ± 0.14**	0.42 ± 0.06	0.68 ± 0.23
Shu et al.[49]	**0.37 ± 0.06**	**0.22 ± 0.03**	0.22 ± 0.11	**0.17 ± 0.03**	**0.52 ± 0.12**
Masked diffusion (ours)	**0.31 ± 0.06**	**0.15 ± 0.02**	**0.15 ± 0.06**	**0.11 ± 0.02**	**0.33 ± 0.07**

1.5% sparse data reconstruction on 2D Kolmogorov data

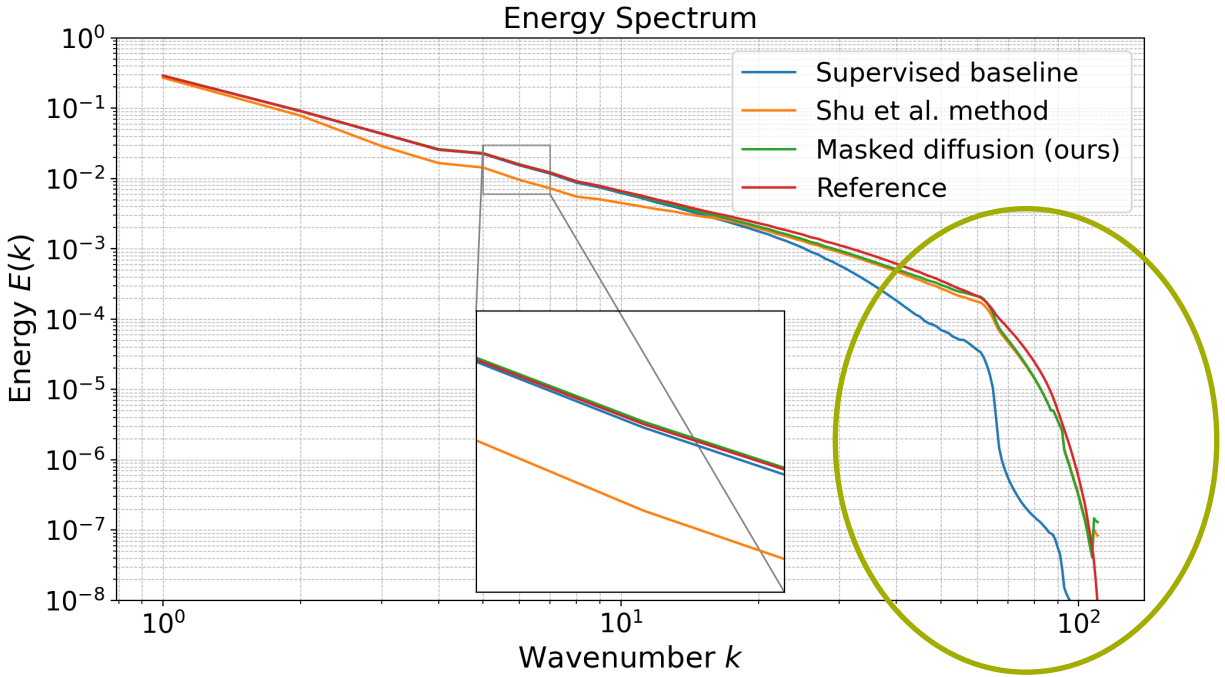
Models	RMSE	P-RMSE	L_{res}	LSiM	E.spectrum
Supervised baseline	**0.16 ± 0.01**	**0.27 ± 0.01**	**0.36 ± 0.01**	**0.08 ± 0.01**	0.71 ± 0.02
Shu et al.[49]	0.44 ± 0.02	0.69 ± 0.04	0.79 ± 0.05	0.62 ± 0.11	**0.36 ± 0.04**
Masked diffusion (ours)	**0.20 ± 0.01**	**0.18 ± 0.01**	**0.66 ± 0.03**	**0.09 ± 0.01**	**0.09 ± 0.01**

1.5% sparse data reconstruction on 3D isotropic turbulence

D. Shu, Z. Li, and A. Barati Farimani. "A physics-informed diffusion model for high-fidelity flow field reconstruction"

Results

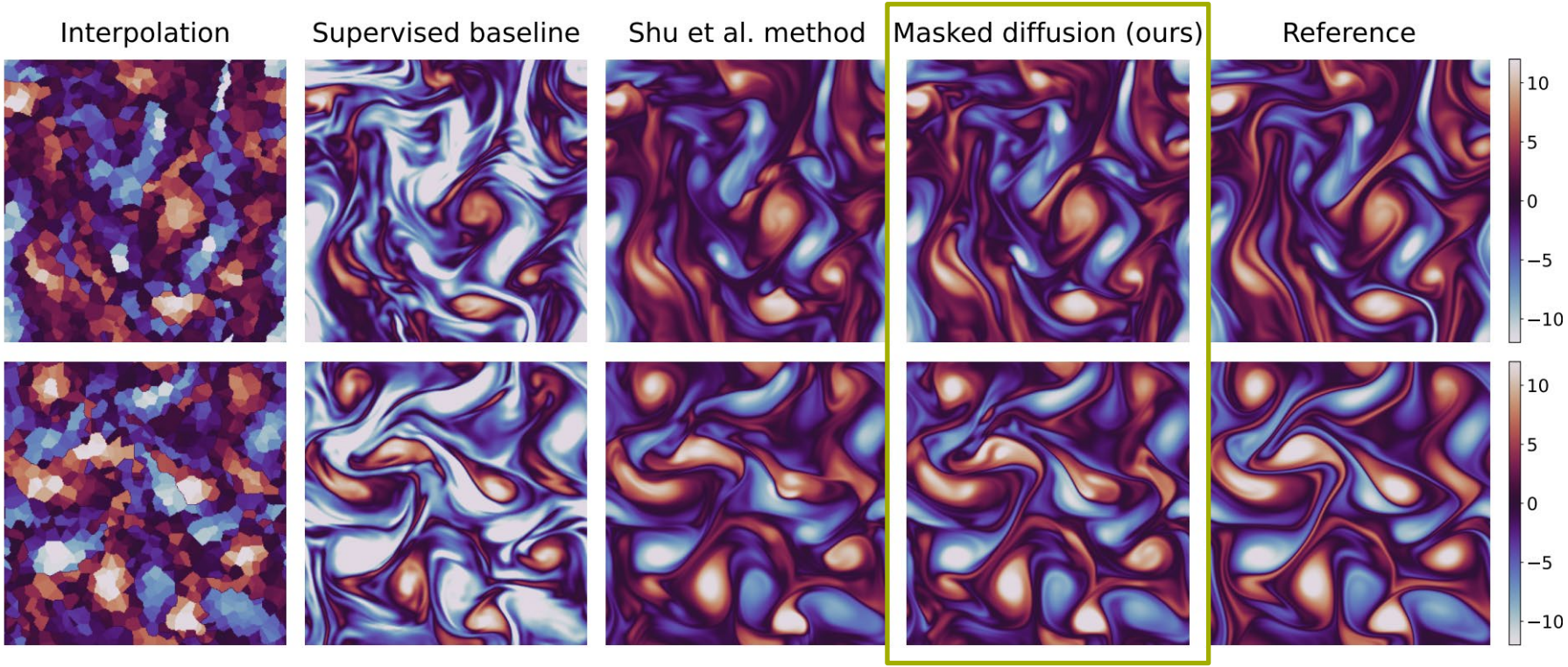
3D isotropic turbulence



1.5% sparse data reconstruction

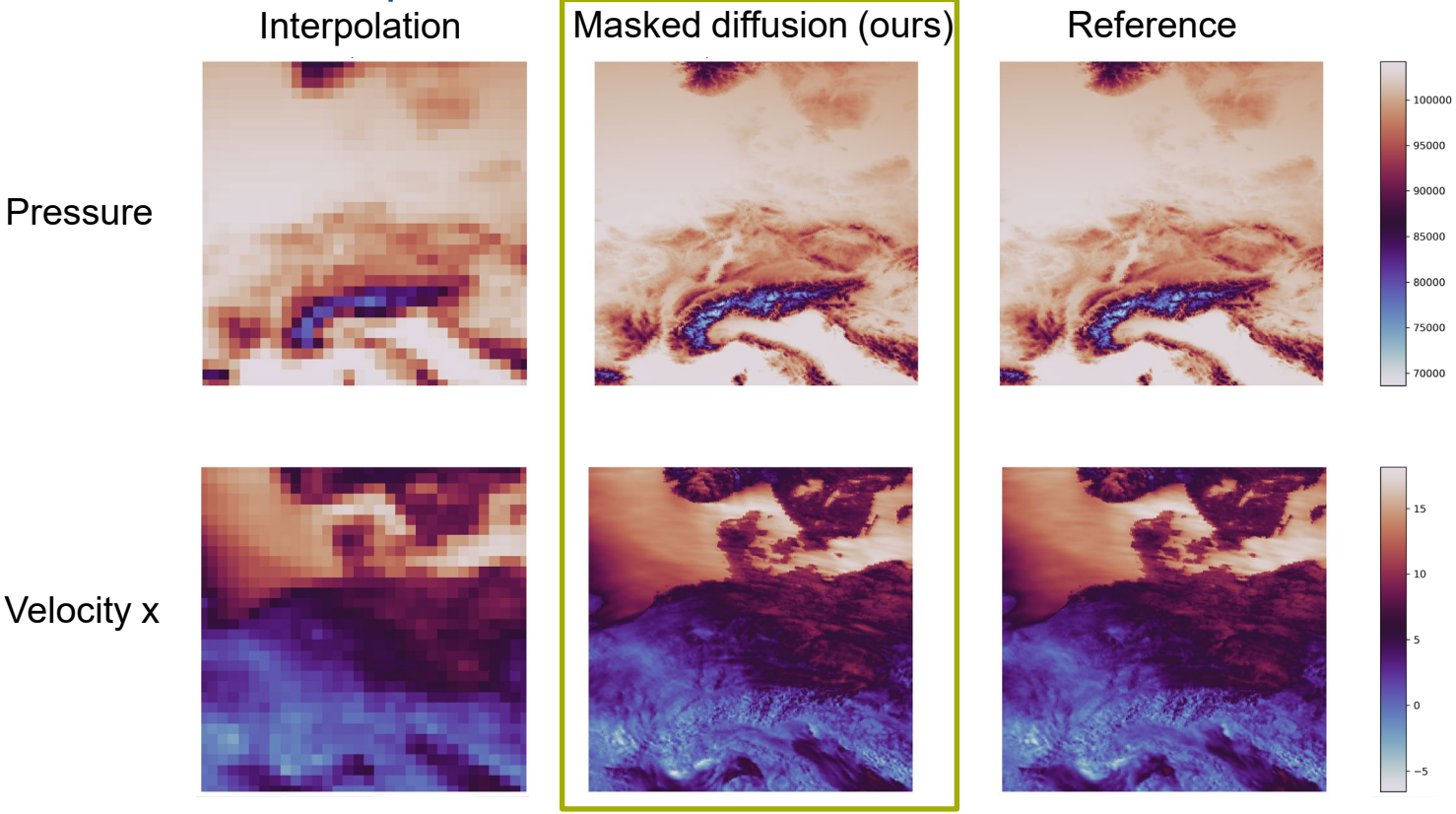
Results

2D Kolmogorov – 1.5% sparse data reconstruction



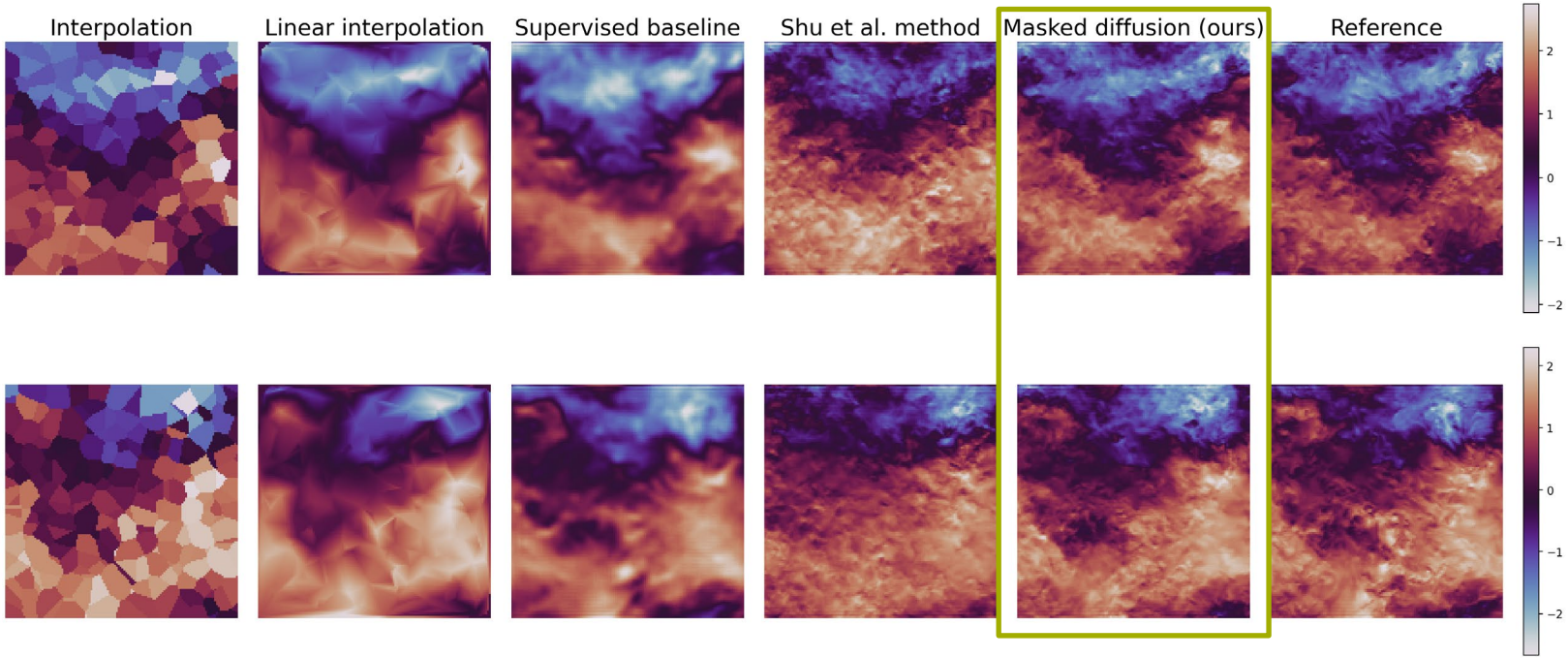
Results

2D Weather – 10x super-resolution



Results

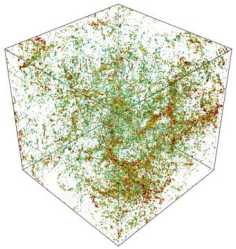
3D isotropic turbulence – 0.1% sparse data reconstruction

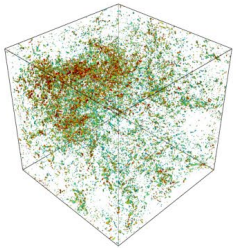


Results

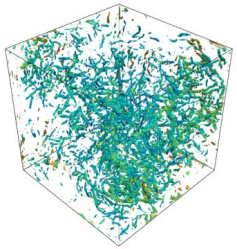
3D isotropic turbulence – Q criterion

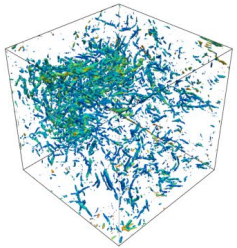
Interpolation



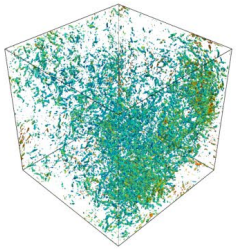


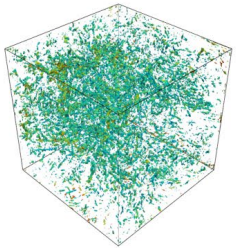
Supervised baseline



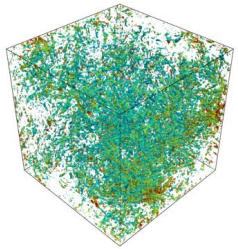


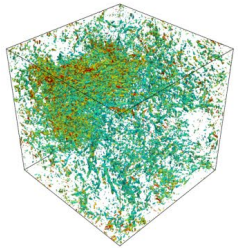
Shu et al. method



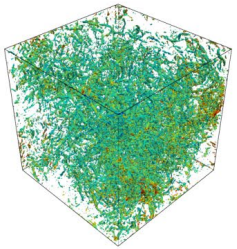


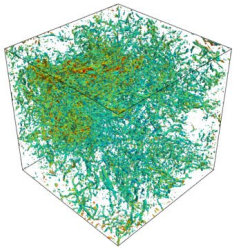
Masked diffusion (ours)





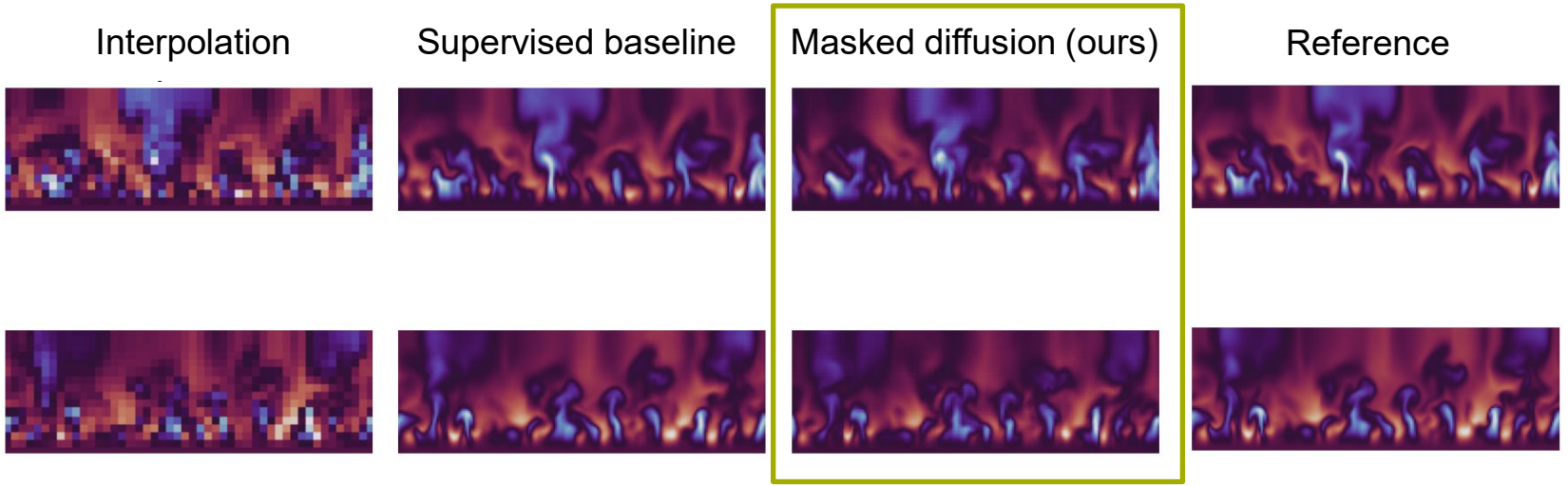
Reference





Results

3D channel flow



Conclusions

- New guidance for masked diffusion
- Physical constraints in diffusion models
- Better reconstruction of small vortices
- Future work
  - Latent diffusion models
  - Other physics or varying Re
  - Uncertainty estimation
  - Noisy experimental data

Thanks for your attention ☺
Any questions?



SCAN ME

References

- M. Amorós-Trepat, L. Medrano-Navarro, Q. Liu, L. Guastoni, and N. Thuerey. “Guiding Diffusion Models to Reconstruct Flow Fields from Sparse Data”. 2026.
- K. Debeire, A. Pacal, P. Gentine, L. Medrano-Navarro, N. Thuerey, and V. Eyring. “Physics-Constrained Adaptive Flow Matching for Climate Downscaling”. 2026.
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